Date	April 27, 2026
Group	70
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Homework	Project
Due	April 28, 2026

Project Description:

In this project we implemented the database handler for an item rental system. Users use a basic command-line interface to interact with the database with entries of rentals, items, customers, and has functionality to search specific tables within the database. While the main file was given, we implemented all the functions to query the database to retrieve and input all the information.

What Works:

- Adding items
- Renting items
- Searching through all tables with filters:
    - Handling for missing address and name fields
    - Handling for min and max price and dates
    - All other filter attributes specified
- Returning items
- Adding customers
- Editing customers
- Granting extensions

What Doesn't Work:

- All functionality in the spec has been implemented

Known Bugs:

- When the same customer rents and returns the same item on the same day, on the second return it throws an error because the primary key is item_id, customer_id and rental_date, so duplicate entries aren't allowed even though they were separate rentals.

Screenshots:

1)

```
(.venv) micah@Micahs-MacBook-Air cis4301sp26_project % python setup_db.py ./tpcds_data/
  Loaded 18000 rows into item
Loading customer_address.csv...
  Loaded 50000 rows into customer_address
Loading customer.csv...
  Loaded 100000 rows into customer
Loading date_dim.csv...
  Loaded 73049 rows into date_dim
Loading store_sales.csv (this may take several minutes)...
  Loaded 2880404 rows into store_sales
Setting i_num_owned to random values (1-5)...
  Updated 18000 items
Populating rental_history from store_sales...
  Inserted 2685933 rows into rental_history

Database setup complete!
Successfully loaded in the data
(.venv) micah@Micahs-MacBook-Air cis4301sp26_project %
```

2)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 1

Enter Item ID: AAAAAAAABAAAAAAA
Enter Customer ID: AAAAAAAABAAAAAAA
Successfully rented item
```

3)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 2

Enter Item ID: AAAAAAAABAAAAAAA
Enter Customer ID: AAAAAAAABAAAAAAA
Successfully returned the item
```

4)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 3

Enter Item ID: AAAAAAAABAAAAAAA
Enter Customer ID: AAAAAAAABAAAAAAA
Successfully granted extension

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 3

Enter Item ID: AAAAAAAABAAAAAAA
Enter Customer ID: AAAAAAAABAAAAAAA
The customer already has an extension and may not be granted another one

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit
```

5)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 1

Would you like to use patterns to search String attributes? (Y/N): n
Which attribute would you like to filter?
1. Item ID
2. Product Name
3. Brand
4. Manufacturer
5. Category
6. Min Price
7. Max Price
8. Min Start Year
9. Max Start Year
10. Continue
11. Cancel

Choice: 3

Brand: importoimporto #2

Current Filters:
--------------------
Brand: importoimporto #2
--------------------
```

```
Current Filters:
--------------------
Brand: importoimporto #2
--------------------

Which attribute would you like to filter?
1. Item ID
2. Product Name
3. Brand
4. Manufacturer
5. Category
6. Min Price
7. Max Price
8. Min Start Year
9. Max Start Year
10. Continue
11. Cancel

Choice: 7

Max Price: 0.5

Current Filters:
--------------------
Brand: importoimporto #2
Max Price: 0.5
--------------------

Which attribute would you like to filter?
1. Item ID
2. Product Name
3. Brand
4. Manufacturer
5. Category
6. Min Price
7. Max Price
8. Min Start Year
9. Max Start Year
10. Continue
11. Cancel

Choice: 10


Current Filters:
--------------------
Brand: importoimporto #2
Max Price: 0.5
--------------------
```

```
---------------------
Item ID: AAAAAAAAKCBAAAAA
Product Name: n stn stable
Brand: importoimporto #2
Category: Men
Manufacturer: antiought
Current Price: 0.26
Start Year: 1999
Total number of copies owned: 5
---------------------
---------------------
Item ID: AAAAAAAAIBFAAAAA
Product Name: antibarpriought
Brand: importoimporto #2
Category: Men
Manufacturer: baroughtanti
Current Price: 0.23
Start Year: 2000
Total number of copies owned: 5
---------------------
---------------------
Item ID: AAAAAAAACNIAAAAA
Product Name: n stantiableable
Brand: importoimporto #2
Category: Men
Manufacturer: n stbarought
Current Price: 0.46
Start Year: 2000
Total number of copies owned: 3
---------------------
---------------------
Item ID: AAAAAAAAAPGBAAAA
Product Name: priationeinganti
Brand: importoimporto #2
Category: Men
Manufacturer: oughteing
Current Price: 0.1
Start Year: 1999
Total number of copies owned: 5
---------------------
---------------------
Item ID: AAAAAAAAKOJCAAAA
Product Name: oughtpriationbarought
Brand: importoimporto #2
Category: Men
Manufacturer: eingbarable
Current Price: 0.1
Start Year: 2000
Total number of copies owned: 2
---------------------
```

```
--------------------
Item ID: AAAAAAAAKOJCAAAA
Product Name: oughtpriationbarought
Brand: importoimporto #2
Category: Men
Manufacturer: eingbarable
Current Price: 0.1
Start Year: 2000
Total number of copies owned: 2
--------------------
--------------------
Item ID: AAAAAAAAALMCAAAA
Product Name: oughteseeseoughtought
Brand: importoimporto #2
Category: Men
Manufacturer: priationought
Current Price: 0.41
Start Year: 1999
Total number of copies owned: 2
--------------------
--------------------
Item ID: AAAAAAAAELCDAAAA
Product Name: oughteingn stableought
Brand: importoimporto #2
Category: Men
Manufacturer: antipriable
Current Price: 0.32
Start Year: 2000
Total number of copies owned: 4
--------------------
--------------------
Item ID: AAAAAAAAAMLDAAAA
Brand: importoimporto #2
Category: Men
Current Price: 0.0
Start Year: 2000
Total number of copies owned: 3
--------------------
--------------------
Item ID: AAAAAAAACDEEAAAA
Product Name: n stantieseationought
Brand: importoimporto #2
Category: Men
Manufacturer: esen stable
Current Price: 0.17
Start Year: 1999
Total number of copies owned: 2
--------------------

Found 9 items.
```

6)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 2

Would you like to use patterns to search String attributes? (Y/N): y
Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 2

Name: %son

Current Filters:
--------------------
Name: %son
--------------------

Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice:
```

```
Found 576 customers.

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice:
```

7)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 3

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Continue
8. Cancel

Choice: 3

Min Rental Date (YYYY-MM-DD): 2026-03-01

Current Filters:
--------------------
Min Rental Date: 2026-03-01
--------------------
```

```
Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Continue
8. Cancel

Choice: 7


Current Filters:
--------------------
Min Rental Date: 2026-03-01
--------------------


--------------------
Item ID: AAAAAAAABAAAAAAA
Customer ID: AAAAAAAABAAAAAAA
Rental Date: 2026-04-27
Due Date: 2026-05-25
--------------------
--------------------
Item ID: TESTITEM02
Customer ID: TESTCUST02
Rental Date: 2026-04-27
Due Date: 2026-05-11
--------------------

Found 2 rentals.

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit
```

8)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 4

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Min Return Date
8. Max Return Date
9. Continue
10. Cancel

Choice: 9
```

```
----------------------
Item ID: AAAAAAAAPPPBAAAA
Customer ID: AAAAAAAAP0GHBAAA
Rental Date: 2000-11-13
Due Date: 2000-11-27
Return Date: 2000-11-27
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----------------------
Item ID: AAAAAAAAPPPBAAAA
Customer ID: AAAAAAAAP0NEAAAA
Rental Date: 1998-05-11
Due Date: 1998-05-25
Return Date: 1998-05-25
----------------------
----------------------
Item ID: TESTITEM01
Customer ID: TESTCUST01
Rental Date: 2026-04-27
Due Date: 2026-05-25
Return Date: 2026-04-27
----------------------
----------------------
Item ID: TESTITEM02
Customer ID: TESTCUST01
Rental Date: 2026-04-27
Due Date: 2026-05-11
Return Date: 2026-04-27
----------------------

Found 2685936 returns.
```

9)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 1

Enter Item ID: AAAAAAAAHAAAAAAA
Enter Customer ID: AAAAAAAABAAAAAAA
Successfully rented item

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 1

Enter Item ID: AAAAAAAAHAAAAAAA
Enter Customer ID: AAAAAAAACAAAAAAA
This item is not available right now.
Would you like to waitlist the Customer (Y/N): y
The customer is now 1st in line to rent the item
```

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 5

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Place in line
4. Max Place in line
5. Continue
6. Cancel

Choice: 1

Item ID: AAAAAAAAHAAAAAAA

Current Filters:
--------------------
Item ID: AAAAAAAAHAAAAAAA
--------------------
```

```
Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Place in line
4. Max Place in line
5. Continue
6. Cancel

Choice: 5


Current Filters:
---------------------
Item ID: AAAAAAAAHAAAAAAA
---------------------

---------------------
Item ID: AAAAAAAAHAAAAAAA
Customer ID: AAAAAAAACAAAAAAA
Place in line: 1
---------------------

Found 1 waitlisted customer.
```

10)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 5

Enter Item ID: test2
Enter Product Name: test2
Enter Brand: test2
Enter Category: test2
Enter Manufacturer: test2
Enter Current Price: 10
Enter Start Year: 2026
Enter the number of copies owned: 1

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 5

Enter Item ID: test2
An item with that ID already exists.
```

11)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 6

Enter Customer ID: 1
Enter First Name: test
Enter Last Name: test
Enter Email: test
Enter Street Number: test
Enter Street Name: test
Enter City: test
Enter State: fl
Enter Zip: 12345

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 6

Enter Customer ID: 1
A customer with that ID already exists.
```

12)

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 7

Customer's Customer ID: 1

Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 2

Name: newName newLast
Current Attributes:
--------------------
Name: newName newLast
--------------------

Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 4

Email: newemail@gmail.com
Current Attributes:
--------------------
Name: newName newLast
Email: newemail@gmail.com
--------------------
```

```
Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 5

Current Attributes:
--------------------
Name: newName newLast
Email: newemail@gmail.com
--------------------


What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 2
```

```
Would you like to use patterns to search String attributes? (Y/N): n
Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 1

Customer ID: 1

Current Filters:
--------------------
Customer ID: 1
--------------------

Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 5


Current Filters:
--------------------
Customer ID: 1
--------------------

--------------------
Customer ID: 1
Name: newName newLast
Address: test test, test, fl 12345
Email: newemail@gmail.com
--------------------

Found 1 customer.
```